

likelihood $p(y|\theta) \propto e^{-\frac{(y-\theta)^2}{2\sigma^2}}$

Prior $p(\theta) \propto e^{A\theta^2 + B\theta + C}$

Aim: this prior is a normal distribution

$$A\theta^2 + B\theta + C = A\left(\theta + \frac{B}{2A}\right)^2 - \frac{B^2}{4A^2} + C$$

$$p(\theta) \propto e^{A\theta^2 + B\theta + C} \propto e^{A\left(\theta + \frac{B}{2A}\right)^2}$$

let $A = -\frac{1}{2\tau_0^2}$

$-\frac{B}{2A} = \mu_0$

Prior

$$p(\theta) \propto e^{-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2} \quad \theta \sim N(\mu_0, \tau_0^2)$$

Aim: Posterior is also normal

$$p(\theta|y) \propto p(\theta) p(y|\theta)$$

$$= e^{-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2} e^{-\frac{(y-\theta)^2}{2\sigma^2}}$$

$$= \exp \left[-\frac{1}{2} \left(\frac{(y-\theta)^2}{\sigma^2} + \frac{(\theta - \mu_0)^2}{\tau_0^2} \right) \right]$$

$$= \exp \left[-\frac{1}{2} \left(\frac{1}{\sigma^2} (y^2 - 2\theta y + \theta^2) + \frac{1}{\tau_0^2} (\theta^2 - 2\theta\mu_0 + \mu_0^2) \right) \right]$$

$$\propto \exp \left[-\frac{1}{2} \left[\left(\frac{1}{\sigma^2} + \frac{1}{\tau_0^2} \right) \theta^2 - 2 \left(\frac{y}{\sigma^2} + \frac{\mu_0}{\tau_0^2} \right) \theta \right] \right]$$

$$= \exp \left[\frac{-1}{2} \left(\frac{1}{\sigma^2} + \frac{1}{\tau_0^2} \right) \left(\theta^2 - 2 \left(\frac{\frac{y}{\sigma^2} + \mu_0/\tau_0^2}{\frac{1}{\sigma^2} + \frac{1}{\tau_0^2}} \right) \theta \right) \right]$$

$$\frac{1}{\tau_1^2} = \frac{1}{\sigma^2} + \frac{1}{\tau_0^2}$$

$$\mu_1 = \frac{\frac{y}{\sigma^2} + \mu_0/\tau_0^2}{\frac{1}{\sigma^2} + \frac{1}{\tau_0^2}}$$

$$\propto \exp \left[\frac{-1}{2\tau_1^2} (\theta - \mu_1)^2 \right]$$

$$\theta|y \sim N(\mu_1, \tau_1^2)$$

$$\begin{aligned}
 p(\tilde{y}|y) &= \int p(\tilde{y}|\theta) p(\theta|y) d\theta \\
 &\propto \int \exp\left(-\frac{(\tilde{y}-\theta)^2}{2\sigma^2}\right) \exp\left(-\frac{(\theta-\mu)^2}{2\tau_1^2}\right) d\theta \\
 &\propto \exp(A\tilde{y}^2 + B\tilde{y} + C)
 \end{aligned}$$

$$\tilde{y}|y \sim N(\cdot, \cdot)$$

$$E(\tilde{y}|y) = E(E(\tilde{y}|\theta, y)|y) = E(E(\tilde{y}|\theta)|y) = E(\theta|y) = \mu_1$$

$$\begin{aligned}
 \text{Var}(\tilde{y}|y) &= E(\text{Var}(\tilde{y}|\theta)|y) + \text{Var}(E(\tilde{y}|\theta)|y) \\
 &= E(\sigma^2|y) + \text{Var}(\theta|y) \\
 &= \sigma^2 + \tau_1^2
 \end{aligned}$$

$$\boxed{\tilde{y}|y \sim N(\mu_1, \sigma^2 + \tau_1^2)}$$

How do we handle multiple data points?

Prior: $p(\theta) \propto \exp\left(-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2\right)$

likelihood: $p(y|\theta) \propto \prod_{i=1}^n \exp\left(-\frac{(y_i - \theta)^2}{2\sigma^2}\right) = \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \theta)^2\right)$

Posterior:

$$p(\theta|y) \propto p(y|\theta)p(\theta)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \theta)^2 - \frac{1}{2\tau_0^2}(\theta - \mu_0)^2\right)$$

$$= \exp\left(-\frac{1}{2\sigma^2} \left(\sum_{i=1}^n y_i^2 - 2\theta \sum_{i=1}^n y_i + n\theta^2\right) - \frac{1}{2\tau_0^2}(\theta^2 - 2\theta\mu_0 + \mu_0^2)\right)$$

Let $\bar{y}$ be the sample mean, so $n\bar{y} = \sum_{i=1}^n y_i$

$$= \exp\left(\theta^2 \left(-\frac{n}{2\sigma^2} - \frac{1}{2\tau_0^2}\right) + \theta \left(\frac{n\bar{y}}{\sigma^2} + \frac{1}{\tau_0^2}\mu_0\right)\right)$$

$$\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)^{-1}$$

$$= \exp\left(-\frac{1}{2\tau_n^2} \left(\theta^2 - 2\tau_n^2 \left(\frac{n\bar{y}}{\sigma^2} + \frac{1}{\tau_0^2}\mu_0\right)\theta\right)\right)$$

$$\mu_n = \tau_n^2 \left(\frac{n\bar{y}}{\sigma^2} + \frac{1}{\tau_0^2}\mu_0\right)$$

$$\propto \exp \left(-\frac{1}{2\tau_n^2} (\theta - \mu_n)^2 \right)$$

$$\theta|y \sim N(\mu_n, \tau_n^2)$$

Example - screws

Data $y = \{5.1, 4.9, 5.2, 5.3, 5.0\}$

$n=5$ $\bar{y} = 5.1$ $\sigma^2 = 0.5$ $\tau_0^2 = 1.0$ $\mu_0 = 5.0$

$$\tau_n^2 = \left(\frac{1}{1} + \frac{5}{0.5} \right)^{-1} = \frac{1}{11} \approx 0.11$$

$$\mu_n = \frac{1}{11} \left(\frac{25.5}{0.5} + \frac{5.0}{1.0} \right) = \frac{1}{11} (51 + 5) = \frac{56}{11} \approx 5.1$$

$$\theta|y \sim N(5.1, 0.1)$$

Review: Scaled inverse χ^2 distribution

$$\theta \sim \text{Inv-}\chi^2(\nu, s^2)$$

$\uparrow$ deg. of freedom
 $\uparrow$ scale

$$p(\theta) = \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} s^\nu \theta^{-(\nu/2+1)} e^{-\frac{\nu s^2}{2\theta}}$$

$$\propto \theta^{-(\nu/2+1)} e^{-\frac{\nu s^2}{2} \cdot \frac{1}{\theta}}$$

Unknown variance σ^2 but known mean θ

likelihood:

$$p(y|\sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y_i - \theta)^2 / 2\sigma^2}$$

$$\propto (\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \theta)^2\right)$$

Define $v := \frac{1}{n} \sum_{i=1}^n (y_i - \theta)^2$ ← sample variance.

$$p(y|\sigma^2) \propto (\sigma^2)^{-n/2} \exp\left(-\frac{nv}{2\sigma^2}\right)$$

Prior:

$$p(\sigma^2) \propto (\sigma^2)^{-(\frac{\nu_0}{2}+1)} \exp\left(-\frac{\nu_0 \sigma_0^2}{2} \cdot \frac{1}{\sigma^2}\right)$$

Posterior:

$$p(\sigma^2|y) \propto p(\sigma^2) p(y|\sigma^2)$$

$$\propto (\sigma^2)^{-(\nu_0/2+1)} \exp\left(-\frac{\nu_0 \sigma_0^2}{2} \cdot \frac{1}{\sigma^2}\right) (\sigma^2)^{-n/2} \exp\left(-\frac{nv}{2\sigma^2}\right)$$

$$\sigma^2 \sim \text{Inv-}\chi^2\left(\nu_0 + n, \frac{\nu_0 \sigma_0^2 + nv}{\nu_0 + n}\right)$$

$$= (\sigma^2)^{-\frac{1}{2}(\nu_0 + n + 2)} \exp\left(-\frac{1}{2\sigma^2} (\nu_0 \sigma_0^2 + nv)\right)$$

Review: Gamma (a, b)

$$P(\theta) = \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta} \\ \propto \theta^{a-1} e^{-b\theta}$$

a - shape
 b - inverse scale

mean: a/b
variance: a/b^2

Poisson Likelihood: $p(y|\theta) = \prod_{i=1}^n \frac{1}{y_i!} \theta^{y_i} e^{-\theta} \propto \theta^{n\bar{y}} e^{-n\theta}$

Prior: $p(\theta) \propto e^{-b_0\theta} \theta^{a_0-1}$

Posterior: $p(\theta|y) \propto p(\theta) p(y|\theta) \propto \theta^{n\bar{y}+a_0-1} e^{-n\theta-b_0\theta}$

$\theta|y \sim \text{Gamma}(a_0 + n\bar{y}, b_0 + n)$

$\frac{3}{200,000}$ died of asthma $\rightarrow 1 \text{ year}$ $\leftarrow$ data
 $\hookrightarrow \frac{1.5}{100,000}$ $x = 2.0$

World data $\frac{0.6}{100,000}$ $\leftarrow$ prior

Prior 0.6 mean $\frac{a}{b} = 0.6$

Suppose $a = 3.0$ and $b = 5.0$

Posterior $\text{Gamma}(a_0 + y, b_0 + x) = \text{Gamma}(6.0, 7.0)$

mean $= \frac{6}{7} \approx 0.86$ variance $= \frac{6}{7^2} < \frac{3}{5^2}$

$\frac{3}{200,000}$ died of asthma $\rightarrow$ 1 year $\leftarrow$ data
 $\hookrightarrow 1.5/100,000$ Exposure $x = 2.0$

World data: $0.6/100,000$ $\leftarrow$ prior

Prior mean 0.6 and a Gamma distribution
 Gamma (α_0, β_0) mean = $\frac{\alpha_0}{\beta_0} = 0.6$

Suppose $\alpha_0 = 3.0$ $\beta_0 = 5.0$

Posterior Gamma ($\alpha_0 + y, \beta_0 + x$) = Gamma (6.0, 7.0)

mean = $\frac{6}{7} \approx 0.86$

Variance $\frac{6}{7^2} < \frac{3}{5^2}$ why?